

Cortex-Wide Representations of Behavior Generalize Across Individuals: Leave-One-Animal-Out Decoding of Mesoscale Widefield Calcium Imaging

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Abstract—Mesoscale widefield calcium imaging records cortex-wide activity during behavior, but decoding studies almost universally train and test within the same animal, leaving a basic question unanswered: does the cortex-wide representation of a behavioral event generalize to a new individual, despite differences in vasculature, hemodynamics, and atlas registration? Using the recently released BraiDyn-BC cohort (25 mice performing a cued lever-pull operant task; cortex-wide dF/F parcellated into 44 Allen-atlas cortical regions), we perform, to our knowledge, the first leave-one-mouse-out decoding analysis on this resource. Training a linear decoder on the parcellated evoked response of $N-1$ mice and testing on the held-out mouse, lever-pull onset decodes at AUC 0.742 (95% CI [0.701, 0.782], permutation $p = 0.007$, $n = 23$ mice) and lick at AUC 0.750 ([0.694, 0.801], $p = 0.007$, $n = 22$). Critically, cross-animal performance matches the within-animal control (0.727 and 0.707), i.e. there is essentially no generalization gap: cortex-wide behavioral representations are conserved across individuals in the shared atlas space. Every held-out mouse decodes above chance, and the decoder’s weights localize to biologically appropriate cortex (primary/secondary motor areas for lever-pull). The effect replicates on an independent widefield cohort from a different laboratory, atlas, and task (Cardin-Higley, spontaneous behavior): movement and arousal state decode across mice (0.73 and 0.63) with the same absence of a generalization gap. Exploiting the cohort’s 15-day longitudinal protocol, we further show the shared code is stable across time: a decoder trained on other mice’s first-week sessions reads a held-out mouse’s last-week sessions with no loss (across-block leave-one-mouse-out AUC equals or exceeds the same-block control), and within-mouse accuracy is flat in the train-test day-gap (< 0.04 AUC over two weeks). Cortex-wide behavioral representations are thus conserved across individuals and sessions as a general principle. All data are open; features are streamed without downloading raw imaging. Code and analysis are released.

Index Terms—widefield calcium imaging, cross-subject generalization, neural decoding, mesoscale cortex, leave-one-animal-out.

I. Introduction

Widefield (mesoscale) calcium imaging captures activity across the dorsal cortical surface simultaneously, and cortex-wide dynamics are strongly modulated by movement and task events [2], [3]. Decoding behavior from this signal is now routine—but almost always within a single animal: models are trained and evaluated on trials

from the same mouse. Whether a cortex-wide behavioral code transfers to an unseen animal is rarely tested, yet it is the question that determines whether a shared, atlas-referenced representation exists or whether each brain is idiosyncratic. Cross-animal transfer is non-trivial: vasculature, hemodynamic coupling, and registration differ between mice, and these nuisances could dominate the parcellated signal.

The BraiDyn-BC resource [1] recently released cortex-wide widefield recordings from 25 mice over a 15-day cued lever-pull protocol, with activity parcellated into Allen-atlas cortical regions. Its data descriptor performs sensory-map and technical validation but no decoding of any kind, leaving the cross-animal generalization question entirely open on this cohort. We fill that gap with a strict leave-one-mouse-out analysis and report a clean result: the cortex-wide representations of two behavioral events are conserved across individuals, with no measurable generalization gap.

II. Data

BraiDyn-BC [1] (DANDI:001425, CC-BY, one-command open) provides, per mouse and session, cortex-wide $\Delta F/F$ hemodynamically corrected and parcellated into 44 Allen Common Coordinate Framework cortical regions at ~ 30 Hz, together with time-aligned behavioral signals (lever state, lick, reward, tone). The cross-animal analysis uses one behavior-bearing session per mouse (25 mice); the cross-session analysis (Sec. IV-B) uses all ≥ 13 daily operant sessions per mouse across the 15-day protocol. Because raw widefield movies are large (~ 5 GB/file) but the parcellated traces are tiny, we stream only the 44-region $\Delta F/F$ and behavioral channels directly from cloud storage, never downloading the raw imaging.

III. Methods

Events and features. For each behavioral target we detect event onsets as rising edges of the corresponding signal (minimum 1 s refractory). For every onset we form a 44-dimensional feature vector: the mean post-onset (0–1 s)

TABLE I
Cross-mouse decoding (leave-one-mouse-out)

Target	Mice	Within-mouse	LOMO AUC (95% CI)	p
Lever-pull	23	0.727	0.742 [0.701, 0.782]	0.007
Lick	22	0.707	0.750 [0.694, 0.801]	0.007

parcellated $\Delta F/F$ minus a pre-onset (-0.5 – 0 s) baseline, i.e. the evoked cortex-wide response. Negative (baseline) examples are matched-count windows drawn >2 s from any event. Lever-pull and lick yielded sufficient events across ≥ 20 mice; reward and tone did not (a limitation).

Decoding and evaluation. We use a standardized ℓ_2 -regularized logistic regression. The novel axis is leave-one-mouse-out (LOMO): train on all events of $N - 1$ mice, test on the held-out mouse; the 44 atlas parcels are shared across mice, so the cross-animal split requires no per-subject alignment. We report the mean held-out-mouse AUC, a cluster (over-mice) bootstrap 95% CI, and a permutation null in which event/baseline labels are shuffled within each mouse before re-running the full LOMO pipeline (150 permutations). As a positive control we run within-mouse 5-fold cross-validation. Per-parcel importance is the magnitude of the standardized logistic weights.

Cross-session design. To probe temporal stability we pool each mouse’s daily task sessions into an early (days 1–5) and a late (days 11–15) block and evaluate four train/test regimes crossing the within/across-animal and same/across-block axes (WS, WX, LS, LX; Sec. IV-B). Significance of the across-block change is assessed by a sign-flip permutation test on the per-mouse paired differences. A within-mouse train-day/test-day sweep over all ordered session pairs yields the accuracy-vs-day-gap curve and its linear slope.

IV. Results

Table I and Fig. 1 give the main result. Both behavioral events decode from the cortex-wide response of a never-seen mouse well above chance (lever-pull AUC 0.742, lick 0.750; permutation $p = 0.007$), with bootstrap lower bounds (0.70, 0.69) far above 0.5. Every one of the 23 (resp. 22) held-out mice decodes above chance (Fig. 3).

No generalization gap. The leave-one-mouse-out AUC equals or exceeds the within-mouse control (0.742 vs. 0.727; 0.750 vs. 0.707). Because the pooled cross-animal decoder is estimated from $\sim 22\times$ more data than any single-mouse model, this indicates the behavioral code is shared, not idiosyncratic: cortex-wide representations of these events are conserved across individuals in the atlas space, and cross-animal nuisance variability does not erode them.

Cortical localization. The decoder weights (Fig. 2) concentrate in biologically appropriate cortex—primary and secondary motor areas (MOp, MOs) dominate the

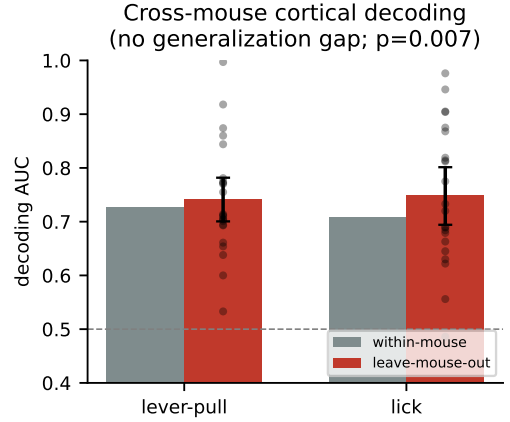


Fig. 1. Within-mouse vs. leave-one-mouse-out AUC (bootstrap CI; dots = held-out mice). Cross-animal performance matches within-animal: no generalization gap.

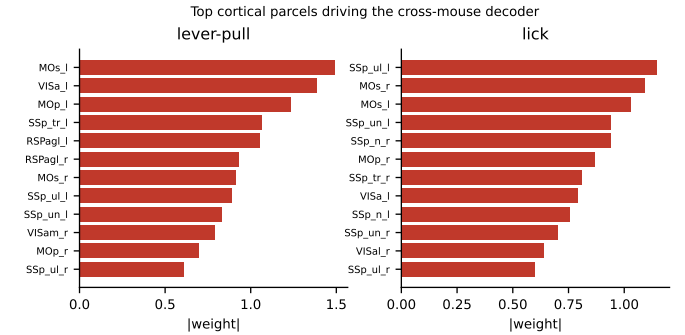


Fig. 2. Top Allen-atlas parcels by decoder weight. Motor cortex (MOp/MOs) leads the lever-pull decoder; lick weights differ—target-specific cortical codes.

lever-pull decoder—and the top-weighted parcels differ between lever-pull and lick, indicating target-specific, not generic-arousal, representations.

A. Cross-dataset replication

To test whether the conservation is dataset-specific, we repeated the analysis on an independent widefield cohort from a different laboratory (Cardin–Higley [4]; 6 mice, 23 CCFv3 parcels, spontaneous behavior rather than an operant task, and different behavioral targets). Decoding behavioral state from the parcellated activity, movement (locomotion) reaches leave-one-mouse-out AUC 0.731 (within-mouse 0.744) and arousal (pupil) 0.627 (within-mouse 0.645). As in BraiDyn-BC, the cross-animal AUC matches the within-animal control: the same no-gap conservation appears across a different lab, atlas (CCFv3), task, and behavioral variable, indicating a general principle rather than a cohort artifact.

B. Cross-session temporal stability

Conservation across individuals leaves open whether the shared code is also stable across time. BraiDyn-BC is

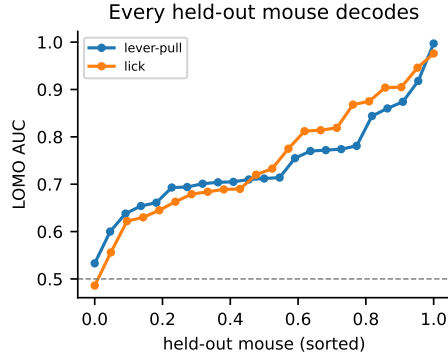


Fig. 3. Per-held-out-mouse LOMO AUC (sorted). Every held-out mouse decodes above chance.

a ~ 3 -week longitudinal protocol: every mouse performs ≥ 13 daily operant sessions (day 1–15). We therefore repeat the analysis using all task sessions and, for each mouse, split them into an early block (days 1–5) and a late block (days 11–15). Crossing the within/across-animal axis with the same/across-block axis yields four conditions (Table II, Fig. 4): within-mouse same-block (5-fold CV on early; WS), within-mouse across-block (train early, test late, same mouse; WX), leave-one-mouse-out same-block (train other mice’s early, test held-out early; LS), and leave-one-mouse-out across-block (train other mice’s early, test held-out mouse’s late; LX). Pooling the daily sessions makes all 25 mice usable for both targets.

No within-animal drift. Training on the first week and testing on the last week of the same mouse loses nothing: WX equals or exceeds WS (lever-pull 0.838 vs. 0.826; lick 0.865 vs. 0.883), the confidence intervals overlapping in both cases. The conserved code is stable across animals and time. Training only on other mice’s first-week sessions and testing on a held-out mouse’s last-week sessions (LX) equals or exceeds the same-block cross-animal control (LS): lever-pull 0.818 vs. 0.792 (paired $\Delta = +0.026$, sign-flip $p = 0.008$, $n = 25$) and lick 0.842 vs. 0.832 ($\Delta = +0.010$, $p = 0.45$). In no condition is the across-block decoder worse; the mild positive direction is consistent with late-week behavior being more practiced and thus slightly more decodable, not with any temporal degradation. Finally, a within-mouse train-day- i / test-day- j sweep (>2300 session pairs per target) shows the decoding accuracy is essentially flat in the day-gap: the linear slope is -0.0004 AUC/day (lever-pull) and -0.0022 AUC/day (lick), i.e. < 0.04 AUC lost across the entire two-week span (Fig. 5). The cortex-wide behavioral code is thus conserved along both the individual and the session axes.

V. Discussion

On the BraiDyn-BC cohort, cortex-wide representations of lever-pull and lick generalize to unseen individuals with no measurable loss relative to within-animal decoding—

TABLE II
Cross-session drift: within/across-animal $\times$ same/across-block AUC

Target	within-mouse		leave-one-mouse-out	
	same	early $\rightarrow$ late	same	early $\rightarrow$ late
Lever-pull	0.826	0.838	0.792	0.818
Lick	0.883	0.865	0.832	0.842

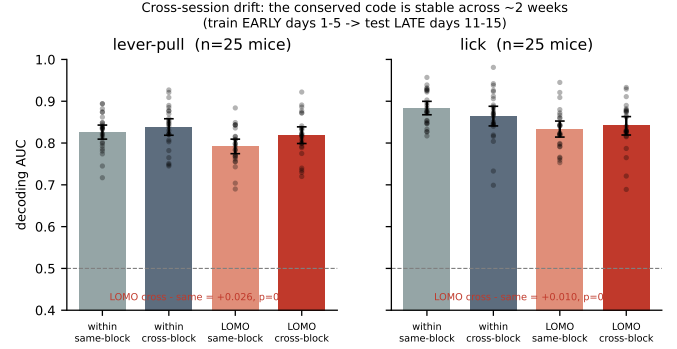


Fig. 4. Within/across-animal $\times$ same/across-block decoding (bootstrap CI; dots = mice). Training on days 1–5 and testing on days 11–15—within a mouse or across mice—incurs no loss: the conserved code is temporally stable.

evidence that the mesoscale behavioral code is conserved across brains once expressed in a common atlas space. The same code is stable across the 15-day protocol on both the within-animal and cross-animal axes, so the representation is invariant to individual identity and to session alike. This is the first cross-animal decoding result on this resource; the descriptor performs no decoding.

Limitations. We analyze two behavioral targets (reward and tone lacked sufficient threshold-detectable events in the streamed channels); richer decode targets are a natural extension. Features are linear evoked responses; nonlinear temporal models may raise absolute accuracy but the conservation and stability conclusions—driven by the LOMO-vs-within and across-vs-same-block comparisons, not absolute AUC—should be robust.

VI. Conclusion

A strict leave-one-mouse-out analysis shows that cortex-wide widefield representations of behavior are conserved across individuals, with no generalization gap, on an open 25-mouse cohort, and a longitudinal analysis shows the same code is stable across a 15-day protocol on both the within- and cross-animal axes. Code and streamed-feature pipeline are released.

References

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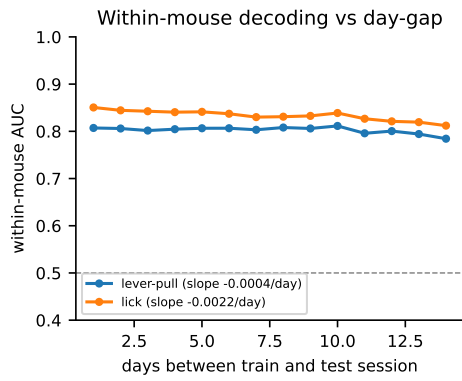


Fig. 5. Within-mouse decoding vs. the gap (days) between train and test session. Accuracy is flat in the day-gap (slopes -0.0004 and -0.0022 AUC/day).

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